

Madhuri Singh

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EDUCATION

Georgia Institute of Technology (Georgia Tech) PhD Student, Computer Science Advisor: Dr. Mark O. Riedl	2025 - Present
Georgia Institute of Technology (Georgia Tech) MS, Computer Science Cumulative GPA: 3.92/4	2022 – 2024
Sardar Vallabhbhai National Institute of Technology (SVNIT) B.Tech, Computer Engineering Cumulative GPA: 9.32/10	2009 - 2013

PEER REVIEWED PUBLICATIONS

[5] Creating Learning Scaffolds for Engineering Design Using Concept Catalyst Madhuri Singh , Gennie Mansi, Mark O Riedl <i>Proceedings of the 6th Annual Meeting of the International Society of the Learning Sciences (ISLS)</i>	2026
[4] Counterfactual Explanations for Agentic Workflows Madhuri Singh , Grace Kim, Mika Okamoto, Aarushi Ammavajjala, Amal Alabdulkarim, Gennie Mansi and Mark O Riedl <i>Proceedings of the CHI 2026 Workshop on Human-Centered Explainable AI (HCXAI)</i>	2026
[3] Explainable Reinforcement Learning Agents Using World Models Madhuri Singh , Amal Alabdulkarim, Gennie Mansi, Mark O Riedl <i>Workshop on Explainable Artificial Intelligence (XAI) at IJCAI 2025.</i>	2025
[2] Experiential Explanations for Reinforcement Learning Amal Alabdulkarim, Madhuri Singh , Gennie Mansi, Kaely Hall, Mark O Riedl. <i>Neural Computing and Applications</i>	2025
[1] Transforming CS Education with DevOps: Streamlined Assignment Validation and Delivery @ Scale Gururaj Deshpande, Shravan Cheekati, Shail Patel, Pranav Raj, Madhuri Singh , Mark Pindur, Nouf Al Soghyar, Bryan Zhao, Parisa Babolhavaeji, Mohammad Taher, Krish Nathan, Will Spaeth and Max Mahdi Roozbahani <i>Proceedings of the Eleventh ACM Conference on Learning@ Scale</i>	2024

ONGOING RESEARCH PROJECTS

Reverse World Models Enabling AI experts and non-experts to change the behavior of their RL agent by altering the environment in specific ways. After proving in [3] that such explanations are possible and useful, we are now creating ways to generate the required action sequences programmatically rather than manually.	2024 – 2025, 2026 – present
Neurodivergent Job Coaching Designing and developing a process to learn human preferences or recommendations obtained by their interactions with an LLM.	2025 – present
Inherently Auditable AI Designing and developing a framework that helps users understand the structure and functioning of an AI pipeline through agentic counterfactual generation, code summarization and fault localization.	2025 – present

After presenting our first design in [4], we continue to explore ways to help people get a better working understanding of their AI systems.

COURSE PROJECTS

Image Deblurring	2024
Compared the strengths and weaknesses of 4 Deblurring models under different image degradation scenarios. [Report]	
Learning Better Truths with Fake Images	2023
Improved the image representations learned by SimCLR on subsets of the ImageNet database, by using data augmentations generated by Diffusion models.	
501 Miles: Piezoelectric Energy Harvesting from Long-Distance Hiking	2023
Developed a prototype of a hiking pole attachment that harvests energy from pole-strikes and stores it in a battery.	
Understanding and Improving Task Transferability in Imitation Learning	2022
Implemented and evaluated task transferability for Behavior Cloning algorithms using a transfer learning algorithm called Side-Tuning. Further implemented the automatic determination of a suitable value for a pivotal hyperparameter.	
Predicting Attrition among Employees	2022
Implemented a neural network based classification model to predict potential employee attrition based on the publicly available Employee Attrition Database.	
Intrusion Detection System in Cloud Environment	2013
Proposed a new design for an Intrusion Detection System which integrated elements of both a Host-based IDS and a Network-based IDS to provide better security for Cloud systems without hurting their performance.	
Sentiment Analysis of Text	2013
Presented a holistic view of the state of research for Sentiment Analysis, cataloging the motivations, various sources, and methods of analysis. Broke down the steps followed to construct, implement, and validate all available approaches for the science.	

EXPERIENCE

Georgia Tech, EI & HCAI Labs	2025 - Present
<i>Graduate Research Assistant (GRA)</i>	
Advised by Dr. Mark O. Riedl	
Areas of focus: Explainable AI, Explainable Reinforcement Learning, Agentic AI	
Georgia Tech, EI & HCAI Labs	2024 – 2025
<i>Research Scientist 1 (RS-1)</i>	
Advised by Dr. Mark O. Riedl	
Areas of focus: Explainable Reinforcement Learning, Artificial Narrow Intelligence	
Georgia Tech, EI & HCAI Labs	2024
<i>Research Affiliate</i>	
Advised by Dr. Mark O. Riedl	
Areas of focus: Explainable Reinforcement Learning	

TEACHING

2026 - 2026	Georgia Tech, School of Interactive Computing
	<i>Graduate Teaching Assistant (GTA)</i>
	Course: AI Storytelling in Virtual Worlds (CS-7634-A)
	Instructor: Dr. Mark Riedl
2023 - 2024	Georgia Tech, School of Interactive Computing
	<i>Graduate Teaching Assistant (GTA)</i>
	Course: Machine Learning (CS 4641/CS 7641)
	Instructor: Dr. Max Mahdi Roozbahani

ACADEMIC SERVICE

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| 2026 | ICLR 2026 2nd Workshop on Deep Generative Model in Machine Learning: Theory, Principle and Efficacy (ICLR 2026 DeLTa Workshop)
<i>Peer Reviewer</i> |
| 2025 | Summit on Responsible Computing, AI, and Society (RCAIS)
<i>Student volunteer</i> |
| 2025 | United States Research Software Engineer Association (US-RSE)
<i>Peer Reviewer</i> |
| 2025 | Practice and Experience in Advanced Research Computing (PEARC)
<i>Peer Reviewer</i> |

INDUSTRY EXPERIENCE

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| 2020 – 2022 | Microsoft IDC
<i>Software Engineer 2</i> <ul style="list-style-type: none">• Designed and implemented a performant and extensible telemetry framework for the OneNote App on Android. Collaborated with the OneNote on iOS team to make sure the learnings are shared and leveraged by partner teams. Also created 50+ dashboards using the Kusto Query Language to automate the querying of telemetry, allowing us to monitor various aspects of the application deployment and its health.• Onboarded new hires with deep enough theoretical and practical understanding of the following topics that they could begin working on them right away.<ul style="list-style-type: none">○ App development on Android○ Microsoft's internal build system○ OneNote Android's development cycle○ Privacy Controls and Telemetry frameworks• Designed and implemented Kotlin based UX (User Experience) workflows for new features added to OneNote Android. The code here involves interactions between an SDK, the Client Android app, and shared C++ code. |
| 2015 – 2019 | Microsoft IDC
<i>Software Engineer</i> <ul style="list-style-type: none">• Investigated, and designed the optimum way to integrate Android Accessibility features with Microsoft's Shared Accessibility framework. Followed this up with expanding the framework's concrete library to help execute various product scenarios for Word Android, Excel Android, Excel iOS and Excel Mac apps.• Designed, prototyped, and implemented OneNote Android's Error Notification System. The work involved the core backend work of fetching & prioritizing error information, as well as development of a reusable UX component to show the error.• Gave Knowledge transfer sessions to multiple teams regarding the following:<ul style="list-style-type: none">○ Microsoft's Shared Accessibility framework○ Android's Accessibility framework○ iOS's Accessibility framework and how it varies from Mac's Accessibility framework○ Error Notification on OneNote Android• Android Q and how to enable Microsoft's Office Android apps to work well with it |
| 2013 – 2014 | Microsoft IDC
<i>Software Development Engineer for Test</i> <p>Created and executed test strategy for multiple fundamental pillars for release of Office Hub app on Android. The areas included the Network layer, Identity and</p> |

- Authentication, Sovereign environments, World Readiness, Knox secure infrastructure (Samsung devices).
- 2012 **Microsoft IDC**
Intern
- Developed an Intelligent Hint Generation to test RoundTrip in Excel and integration with Distributed framework to enable large scale RoundTrip Testing runs.
 - Development of Syntax Checking tool for JavaScript files.
- 2011 **TATA Teleservices Limited**
Intern
- Developed macro-based Automatic report management code for Printer usage, for all TTSL headquarters in Gujarat.

SKILLS

Technologies

DSPy, MLFlow, JAX, Slurm, VS Code, Vue, Anaconda

Programming Languages

Python, C++, C, Kotlin, Javascript

Areas of expertise

Training RL models for sequential decision making using model based (DreamerV3) or model-free (PPO) techniques; Creating custom RL training environments based on Crafter, along with custom reward strategies; Editing World models (DreamerV3); Training supervised and unsupervised models to work with image or tabular datasets; Designing and implementing Qualtrics based human experiments, followed by performing quantitative and qualitative analysis of the responses;

REFERENCES

Dr. Mark O. Riedl

Professor of Interactive Computing, Georgia Tech

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